

# Parameter Estimation and Optimal Control in Epidemic Modelling

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# Introduction

The COVID-19 pandemic emerged in late 2019 and spread globally within months, becoming one of the most significant public health crises of the modern era.

Mathematical epidemic models provide a structured way to understand how diseases spread, estimate key transmission parameters, and evaluate the potential impact of interventions.

The SIR framework — introduced by Kermack and McKendrick (1927) — remains the foundational tool in epidemiology nearly a century later due to its tractability and interpretability.

**670 M+**

Confirmed Cases (Global)

**6.9 M**

Confirmed Deaths (Global)

**~1.03%**

Case Fatality Rate

**3.5 M/day**

Peak Daily Cases (Jan 2022)

# Objectives

- 1 Study the spread of infectious diseases using the SIR epidemic model
- 2 Understand the roles of susceptible, infected, and recovered populations
- 3 Estimate key model parameters: transmission rate  $\beta$  and recovery rate  $\gamma$
- 4 Compare model predictions with observed epidemic data from JHU CSSE
- 5 Analyse how changes in parameters affect the spread of disease
- 6 Study the effect of intervention strategies such as vaccination or social distancing
- 7 Compare epidemic dynamics with and without control measures
- 8 Understand how mathematical modelling supports public health decision-making

# Data Collection

## Johns Hopkins University CSSE Repository

The JHU CSSE COVID-19 dataset is one of the most comprehensive and widely cited open-access pandemic data sources. It was maintained from January 2020 until March 2023.

### Files used:

- CONVENIENT\_global\_confirmed\_cases.csv
- CONVENIENT\_global\_deaths.csv

Format: each column = country/region; each row = calendar date; values = daily new counts (not cumulative).

1,143

Days Covered

Jan

201

Countries & Territories

Mar

2020

Data Start

2023

Data End

~670 M

Total Cases

~6.9 M

Total Deaths

# Data: Important Caveats

## Undercounting

Confirmed case counts significantly underestimate true infections. Testing capacity varied enormously across countries and over time — especially during early waves.

## Deaths Reliability

Death counts are generally more reliable than case counts but are still subject to reporting delays and definitional differences between countries.

## Negative Values

The raw data occasionally contains negative daily counts due to retrospective reporting corrections. These were clipped to zero during preprocessing.

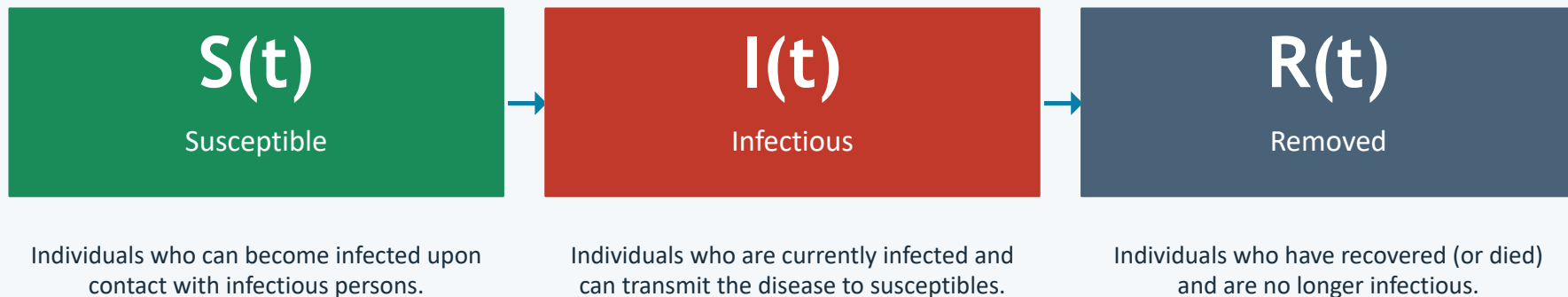
## Sub-national Regions

The dataset includes sub-national regions (e.g., Australia.1 for New South Wales). These were aggregated back into national totals during preprocessing.

## Rich-country Bias

High-income countries dominate the top-10 rankings partly due to superior testing infrastructure, not necessarily higher true infection burden.

# Mathematical Modelling: The SIR Framework



$$\text{Conservation Law: } N = S(t) + I(t) + R(t) = \text{constant}$$

## Key Assumptions:

- Homogeneous, well-mixed population — every individual can contact any other
- No births, deaths from other causes, or migration (closed population)
- Recovery confers permanent immunity — once in R, individuals do not become susceptible again

# Mathematical Modelling: SIR Differential Equations

## System of ODEs

$$\frac{dS}{dt} = -\beta \cdot S \cdot I$$

$$\frac{dI}{dt} = \beta \cdot S \cdot I - \gamma \cdot I$$

$$\frac{dR}{dt} = \gamma \cdot I$$

## Parameter Definitions

$\beta$

Transmission rate (contacts  $\times$  infection prob per day)

$\gamma$

Recovery rate ( $1/\gamma$  = mean infectious period)

$N$

Total (constant) population size

$S \cdot I$

Rate of new infections per day

$\gamma \cdot I$

Rate of recoveries per day

Basic Reproduction Number:

$$R_0 = \beta / \gamma$$

$R_0 > 1$ : spreads |  $R_0 = 1$ : stable |  $R_0 < 1$ : dies out



# Mathematical Modelling: Why Numerical Methods?

The SIR system has no closed-form solution in general.

The nonlinear term  $\beta SI$  makes the system non-separable in time. Dividing  $dI/dt$  by  $dS/dt$  gives a relationship in the phase plane:

$$I(S) = N - S + (\gamma/\beta) \cdot \ln(S/S_0)$$

This gives the shape of the epidemic trajectory but NOT  $S(t)$  explicitly. The resulting ODE for  $S(t)$  contains a logarithm and is not analytically integrable — even Kermack & McKendrick could not write a closed-form  $S(t)$  in their 1927 paper.

## Runge-Kutta 4th Order (RK4)

$$\mathbf{y}_{n+1} = \mathbf{y}_n + (h/6) (\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4)$$

Global error  $O(h^4)$  — highly accurate for epidemic trajectories.  
Step size  $h = 1$  day used throughout.

## SciPy odeint / LSODA

- Automatically switches between Adams (non-stiff) and BDF (stiff) solvers
- Adapts as epidemic stiffness changes over time
- Used in the actual Python implementation

# Mathematical Modelling: Parameter Estimation

## Minimising Sum of Squared Residuals (Nonlinear Least Squares)

Since the SIR model predicts cumulative cases as  $I(t) + R(t)$ , and observed case counts  $y_i$  differ from model predictions  $\hat{y}_i$  by Gaussian noise, maximising the likelihood is equivalent to minimising:

$$SSR(\beta, \gamma) = \sum_i (y_i - \hat{y}_i(\beta, \gamma))^2$$

|                       |                                                                                                               |
|-----------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Initial Guess:</b> | $\beta_0 = 0.3, \gamma_0 = 0.1$ — physically plausible starting point                                         |
| <b>Bounds:</b>        | Both $\beta, \gamma \in (10^{-4}, 10)$ — prevents degenerate solutions                                        |
| <b>Algorithm:</b>     | L-BFGS-B: Limited-memory Broyden–Fletcher–Goldfarb–Shanno with box constraints                                |
| <b>Evaluation:</b>    | At each iteration, odeint integrates the SIR system forward and compares with data                            |
| <b>Convergence:</b>   | <b>Fitted: <math>\hat{\beta} = 5.0000, \hat{\gamma} = 4.9685</math>, giving <math>R_0 \approx 1.01</math></b> |

# Mathematical Modelling: Epidemic Control

Modified SIR with Control Variable  $u(t)$

$$dI/dt = \beta(1 - u(t)) \cdot S \cdot I - \gamma \cdot I$$

$u(t) \in [0, 1]$  — fractional reduction in contact rate due to NPIs, masking, or vaccination

$$R_{eff}(t) = \beta(1 - u) \cdot S(t) / (\gamma \cdot N)$$

$R_0 \approx 1.01$  (Global Fit)

**Required control:  $u > 0.01$**

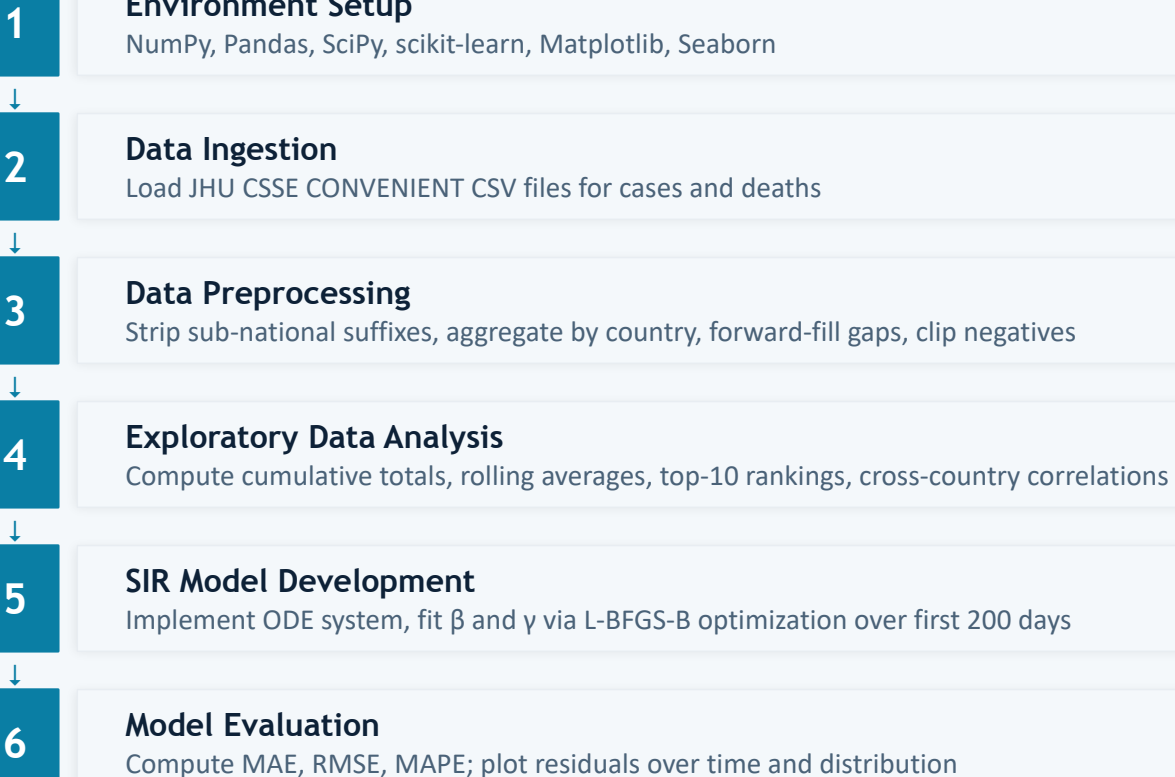
Even a 1% reduction in contacts suppresses the epidemic

$R_0 \approx 3.0$  (Country Level)

**Required control:  $u > 0.67$**

A 67% reduction is required — consistent with strict lockdown findings

# Methodology: Analysis Pipeline



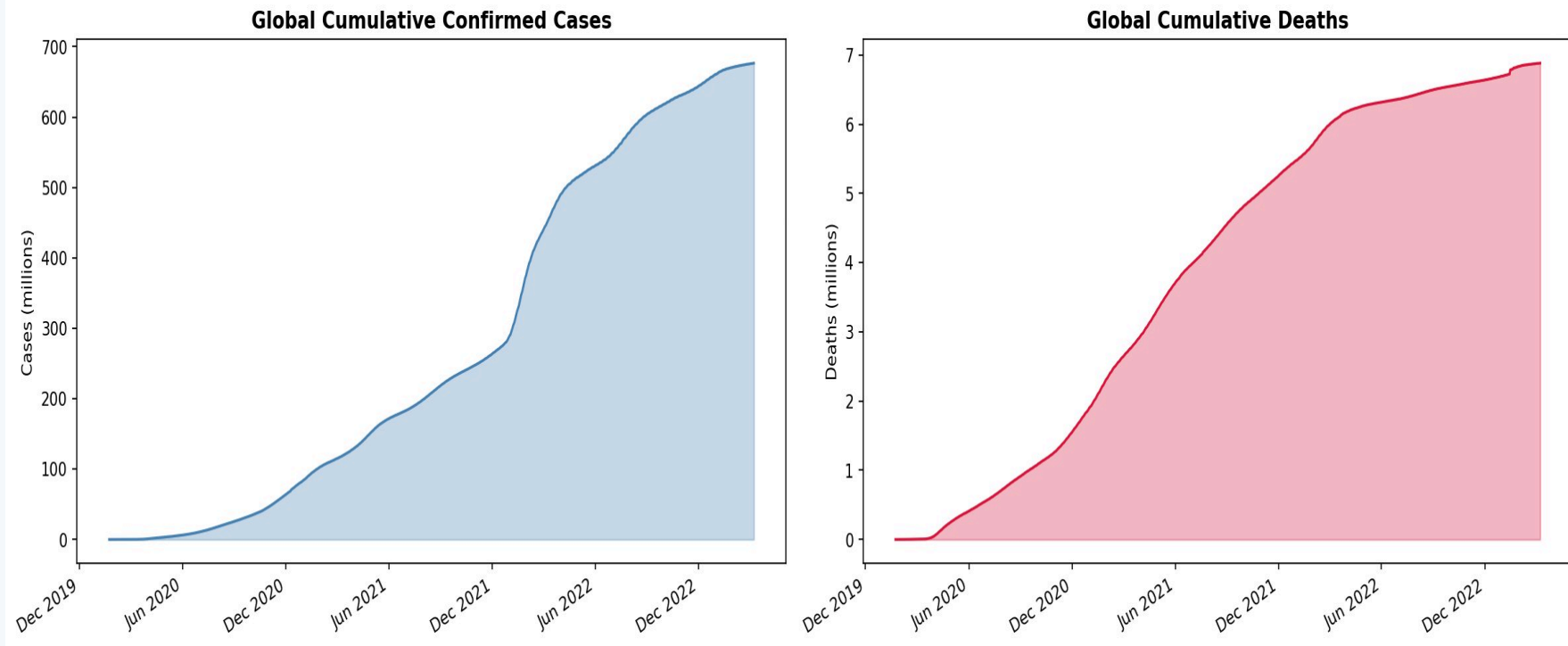
# Methodology: Data Preprocessing

```
def preprocess(df):  
    dates_col = df.iloc[1:, 0].copy()  
    data_part = df.iloc[1:, 1:].copy()  
    # Strip sub-national suffixes: 'Australia.1' → 'Australia'  
    countries = [c.split('.')[0].strip() for c in df.columns[1:]]  
    data_part.columns = countries  
    data_part = data_part.apply(pd.to_numeric, errors='coerce')  
    # Aggregate sub-national regions into parent country  
    data_part = data_part.T.groupby(level=0).sum().T  
    data_part['Date'] = pd.to_datetime(dates_col.values, format='%m/%d/%y')  
    data_part = data_part.sort_values('Date').reset_index(drop=True)  
    data_part = data_part.ffill().dropna(axis=1, how='all')  
    return data_part
```

## Key preprocessing steps:

Remove metadata header · Strip sub-national suffixes · Aggregate by country · Forward-fill sparse gaps · Clip negatives to zero

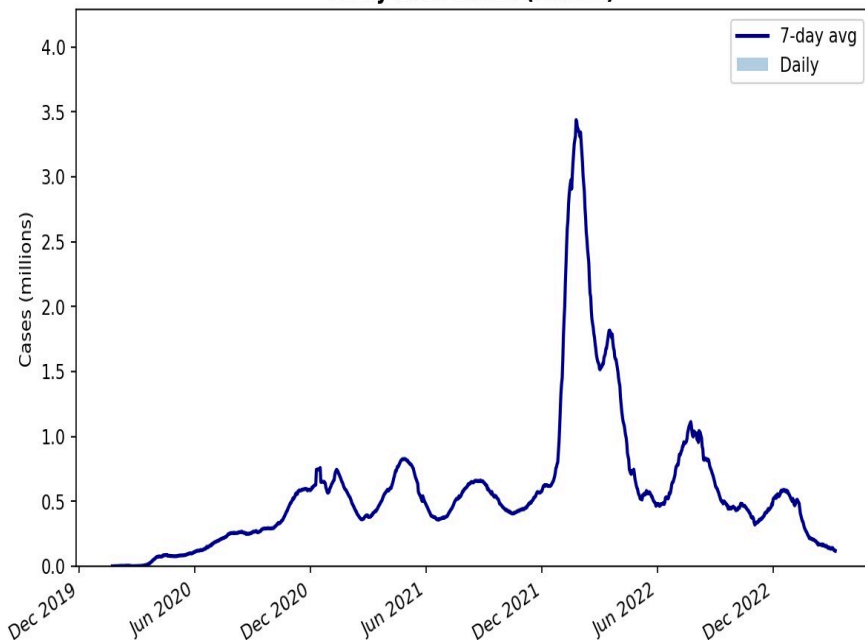
# Results: Global Cumulative Trends



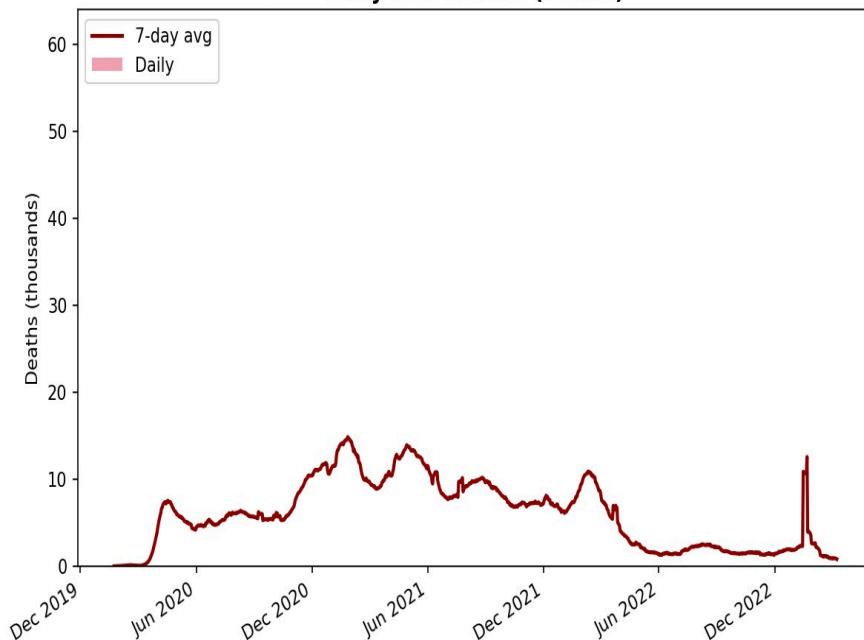
**Cumulative cases reached ~670M by early 2023.** Multiple growth phases correspond to successive variant-driven waves. Deaths (~6.9M) grew more gradually, reflecting improving clinical management and vaccination over time.

# Results: Daily New Cases & Deaths (Global)

Daily New Cases (Global)



Daily New Deaths (Global)



Jan 2022: Omicron peak ~3.5M cases/day — largest single-day surge of the pandemic

Late 2021: Delta wave drove the highest death toll — ~15K deaths/day globally

# Results: Four Epidemic Waves

## Wave 1

Early 2020

Initial global spread following emergence from Wuhan. Spread driven by international travel before containment measures. First lockdowns imposed globally.

## Wave 2

Late 2020

Alpha variant emerged in UK; spread across Europe and Americas during winter. Healthcare systems under severe strain. Vaccine rollout began December 2020.

## Wave 3

Mid 2021

Delta variant ( $R_0 \approx 5-6$ ) drove surges globally, including major waves in India and Southeast Asia. Higher severity than Omicron.

## Wave 4

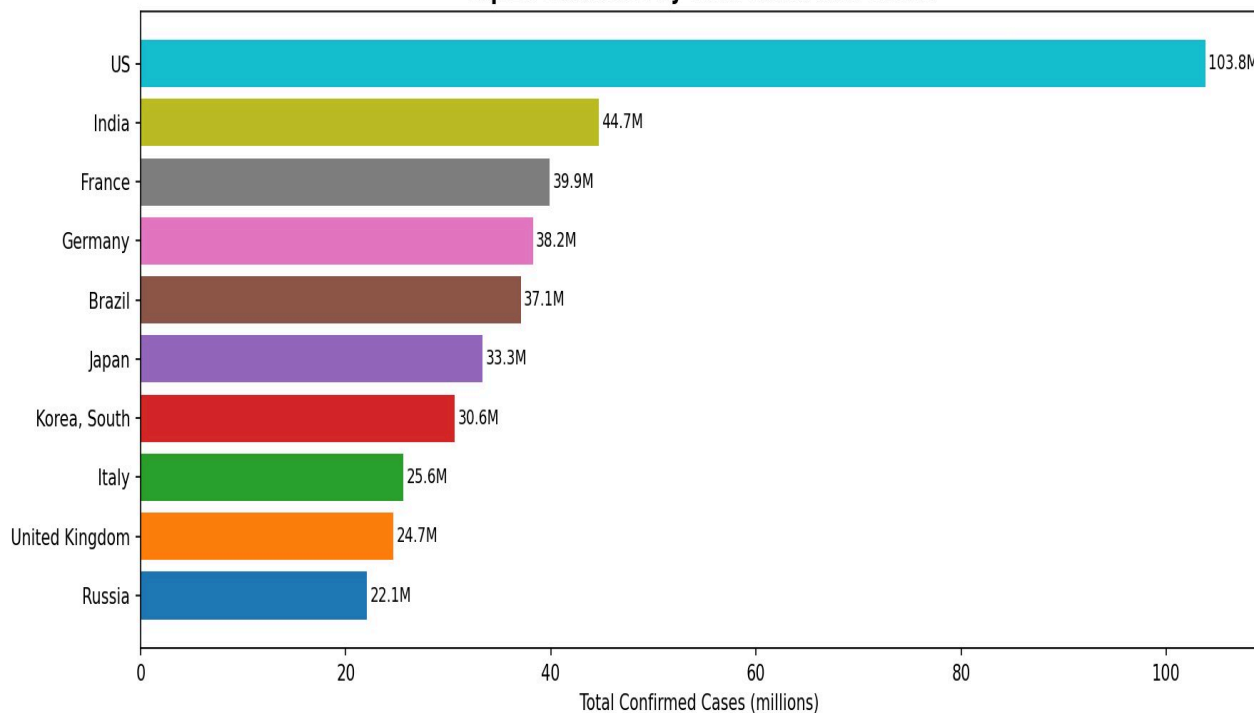
Jan 2022

Omicron ( $R_0 \approx 8-15$ ) produced the largest single-day case counts ever ( $\sim 3.5\text{M/day}$ ) but lower mortality per case due to reduced severity and high vaccination rates.



# Results: Top 10 Countries by Total Confirmed Cases

Top 10 Countries by Total Confirmed Cases



## US leads globally

103.8M confirmed cases — nearly 2.3× India's count

## Testing bias

High-income countries dominate due to superior testing infrastructure

## True burden

LMIC countries likely had higher actual infections but less diagnostic capacity

## CFR variation

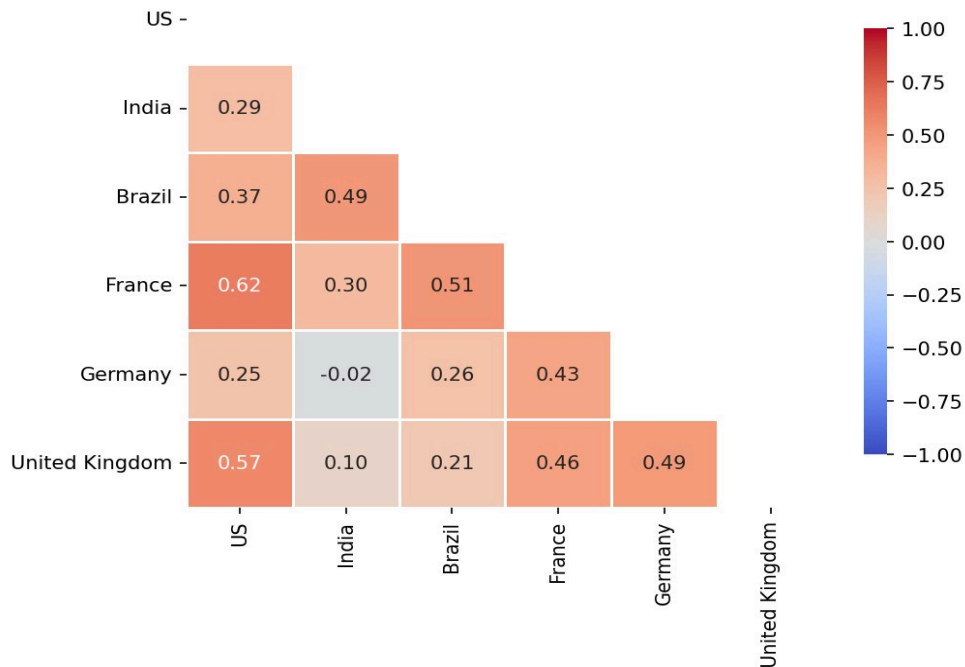
CFR ranged from <0.1% (some countries) to >3% depending on healthcare capacity

# Results: Top 10 Countries – Data Summary

| Country        | Cases (M) | % of Global Total |
|----------------|-----------|-------------------|
| United States  | 103.8     | 15.5%             |
| India          | 44.7      | 6.7%              |
| France         | 39.9      | 6.0%              |
| Germany        | 38.2      | 5.7%              |
| Brazil         | 37.1      | 5.5%              |
| Japan          | 33.3      | 5.0%              |
| Korea, South   | 30.6      | 4.6%              |
| Italy          | 25.6      | 3.8%              |
| United Kingdom | 24.7      | 3.7%              |
| Russia         | 22.1      | 3.3%              |

# Results: Cross-Country Correlation of Daily Cases

Correlation of Daily New Cases  
(Major Countries)



## Key Findings

**0.62**

**US ↔ France**

Strongest correlation — similar epidemic timing

**0.57**

**US ↔ UK**

Shared variant introduction pathways

**0.49**

**Brazil ↔ India**

Moderate synchrony

**-0.02**

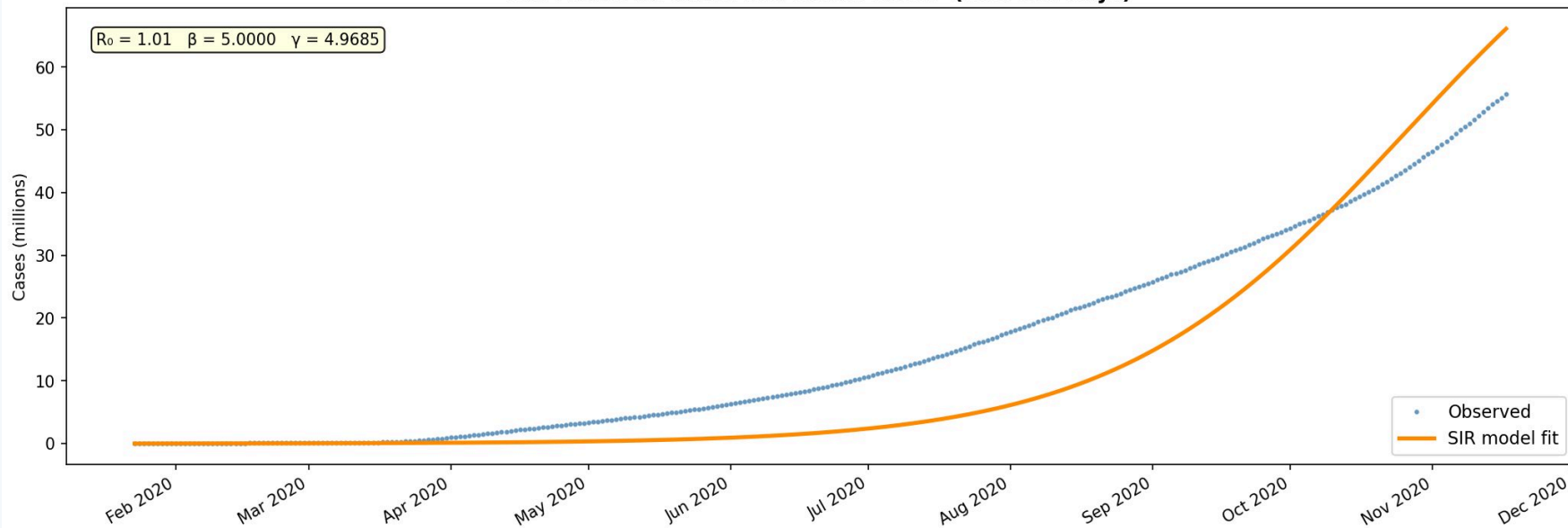
**India ↔ Germany**

Near-zero — distinct dynamics

*Moderate correlations (0.21–0.62) suggest partial global synchrony — international travel seeds variants, but local factors*

# Results: SIR Model Fit (First 300 Days)

SIR Model vs. Observed Global Cases (first 300 days)



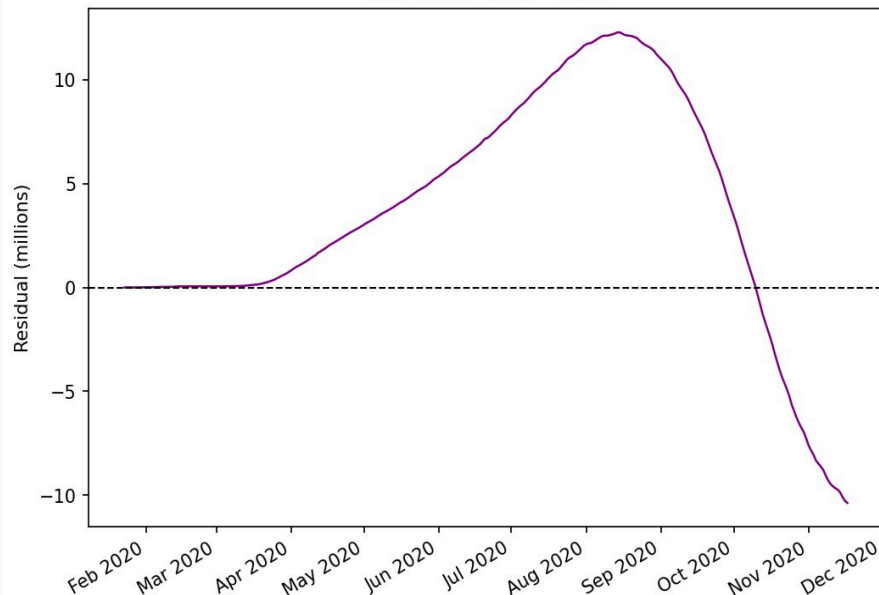
5.0000  
 $\beta$

4.9685  
 $\hat{\gamma}$

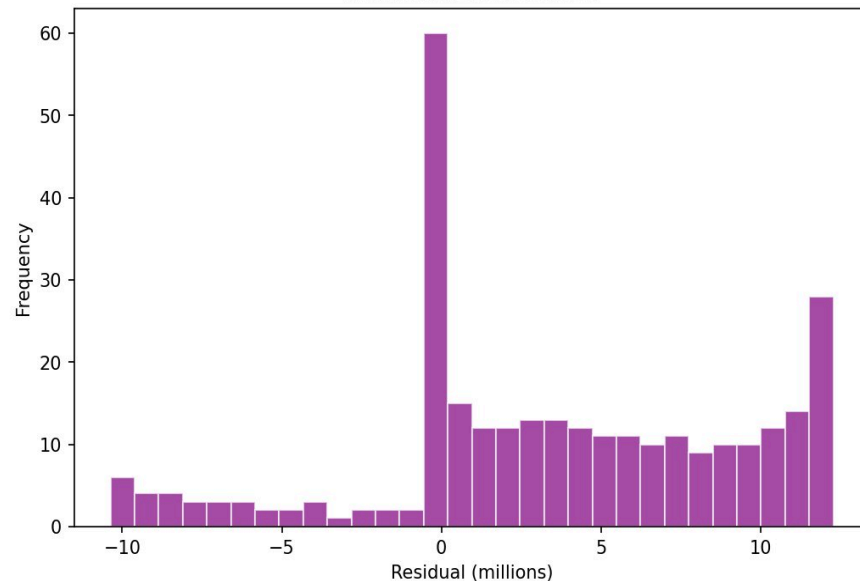
$\approx 1.01$   
 $R_0$

# Results: Residual Analysis

Residuals Over Time



Residual Distribution



## Apr-Sep 2020

Positive residuals (+12M peak): model underestimates as pandemic spreads to new regions faster than a single-wave SIR can predict

## Oct-Dec 2020

Negative residuals (-10M): model overestimates as NPIs and early vaccination slow actual transmission

# Results: Model Evaluation Metrics

**MAE**

~4.2 M cases

Mean Absolute Error — average absolute deviation between predicted and observed cumulative cases

**MSE**

~33.6 M<sup>2</sup>

Mean Squared Error — heavily penalises large deviations; useful for detecting outlier residuals

**RMSE**

~5.8 M cases

Root Mean Squared Error — same units as cases; directly interpretable

**MAPE**

~17.2%

Mean Absolute Percentage Error — scale-free metric; (+1 guard against zero division on early days)

*Bimodal residual distribution (not centred at zero) confirms model misspecification — the basic SIR cannot capture multi-wave pandemic dynamics*

# Discussion: Exploratory Findings

## Multi-wave Structure

Four distinct surges are visible in the daily case data: early 2020 (initial spread), late 2020 (Alpha), mid-2021 (Delta), and January 2022 (Omicron). Each wave reflects a different variant's transmissibility profile.

## Omicron Paradox

Omicron produced by far the highest case counts (~3.5M/day) but comparatively fewer deaths per case — reflecting reduced intrinsic severity and widespread population immunity from prior infection and vaccination.

## Rich-country Bias

High-income countries dominate the top-10 case rankings partly due to superior testing infrastructure. Low- and middle-income countries almost certainly had higher absolute infection numbers but lacked diagnostic capacity.

## Partial Global Synchrony

Cross-country correlations are moderate (0.21–0.62), not high. International travel seeded variants globally, but local factors (vaccine uptake, NPIs, prior immunity, healthcare capacity) shaped each country's specific surge profile.

# Discussion: Model Performance & $R_0$ Discrepancy

## Why is the fitted $R_0 \approx 1.01$ so much lower than published estimates (2.2-3.6)?

The basic SIR assumes a homogeneous, well-mixed population with constant transmission and recovery rates. In reality, the global aggregate data blends 201 countries with heterogeneous populations, time-varying government interventions, immunity accumulation, successive variant replacements, and large cross-country testing differences. The fitted  $\beta$  and  $\gamma$  are effective averages across all these sources of variation — not true biological parameters.

### ⚠ Homogeneity

Assumes well-mixed population — not valid globally across 201 diverse countries

### ⚠ Single wave

Predicts one peak; COVID-19 had four distinct waves driven by different variants

### ⚠ Underreporting

Confirmed cases undercount true infections — biases parameter estimates

### ⚠ Constant rates

$\beta$  and  $\gamma$  cannot vary over time — can't capture changing variants or interventions

### ⚠ Global averaging

Fitting global aggregate data averages out heterogeneous country-level dynamics

### ⚠ No immunity waning

Assumes permanent immunity — not realistic for COVID-19 over 3+ years



# Discussion: Natural Extensions to the SIR Model

**SIR**

$S \rightarrow I \rightarrow R$

Basic model: constant rates, one wave, homogeneous mixing

**SEIR**

$S \rightarrow E \rightarrow I \rightarrow R$

Adds Exposed compartment for incubation period  
(COVID-19: ~5 days)

**SEIRV**

$S \rightarrow E \rightarrow I \rightarrow R$   
 $S \rightarrow V$

Adds Vaccinated compartment — captures immunity from vaccination

**Time-  
varying  
 $\beta(t)$**

$\beta = \beta(t)$

Fits  $\beta$  as piecewise or spline function — absorbs changing interventions and variants

*Optimal control via Pontryagin's Maximum Principle: formal derivation of  $u^*(t)$  that minimises a cost functional balancing total infections against economic cost of intervention*

# Conclusion

## Multi-wave pandemic

Global COVID-19 exhibited ~670M cases and ~6.9M deaths with four distinct waves. The Omicron surge (Jan 2022) produced the largest single-day case peak (~3.5M/day).

## Cross-country correlation

Pearson correlations ranged from  $-0.02$  (India–Germany) to  $0.62$  (US–France), indicating partial global synchrony with substantial local heterogeneity.

## Control variable

A control  $u(t)$  reduces  $R_{\text{eff}}$  proportionally. Suppression requires  $u > 0.01$  for  $R_0 \approx 1.01$ , and  $u > 0.67$  for the more realistic  $R_0 \approx 3$  — consistent with empirical lockdown studies.

## Country distribution

The US reported the highest total (103.8M). High-income country dominance reflects testing capacity as much as true infection burden.

## SIR parameters

Fitted  $\hat{\beta} = 5.0000$ ,  $\hat{\gamma} = 4.9685$ ,  $R_0 \approx 1.01$ . Residual analysis revealed systematic non-random misfit attributable to multi-wave dynamics not capturable by a single-wave SIR.

## Broader value

Despite simplifying assumptions, the SIR framework remains a tractable and interpretable foundation for public health decision-making and epidemic modelling.

# Thank You!

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## Questions & Discussion

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